

INMO(12th) – 1997

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let ABCD be a parallelogram. Suppose a line passing through C and lying outside the parallelogram meets AB and AD produced at E and F respectively. Show that $AC^2 + CE.CF = AB.AE + AD.AF$.
 2. Show that there do not exist positive integers m and n such that $\frac{m}{n} + \frac{m+1}{n} = 4$.
 3. If a, b, c are three distinct real numbers and $a + \frac{1}{b} = b + \frac{1}{c} = c + \frac{1}{a} = t$ for some real number t, prove that $abc + t = 0$.
 4. In a unit square one hundred segments are drawn from the centre to the sides dividing the square into one hundred parts (triangle and possibly quadrilaterals). If all the parts have equal perimeter p, show that $1.4 < p < 1.5$.
 5. Find the number of 4X4 arrays whose entries are from the set {0, 1, 2, 3} and which are such that the sum of numbers in each of the four rows and each of the four columns is divisible by 4. (An mXn array is an arrangement of mn numbers in m rows and n columns.)
 6. Suppose a and b are two positive real numbers such that the roots of the cubic equation $x^3 - ax + b = 0$ are all real. If α is a root of this cubic with minimum absolute value, prove that $\frac{b}{a} < \alpha \leq \frac{3b}{2a}$.

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